Hercules Against Data Series Similarity Search

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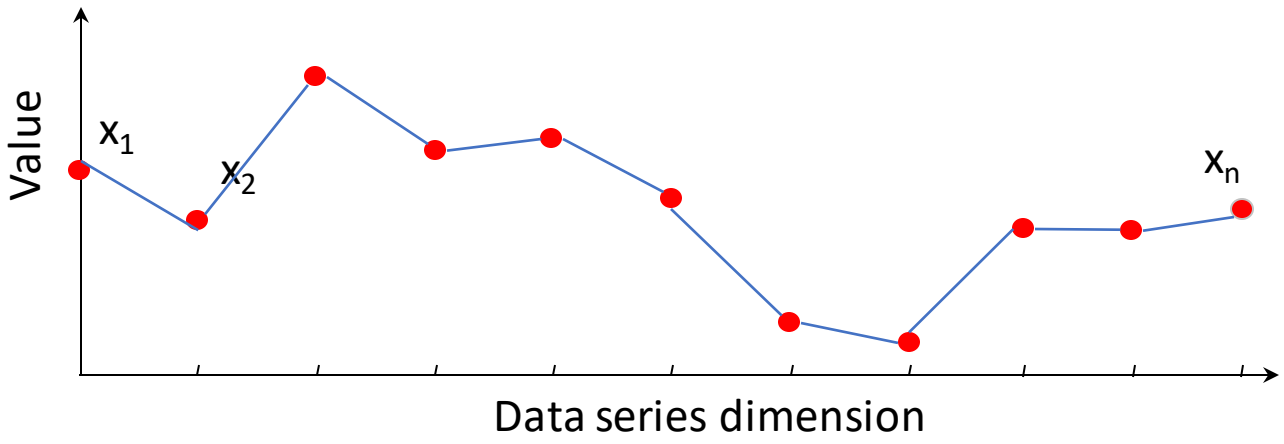
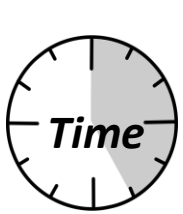
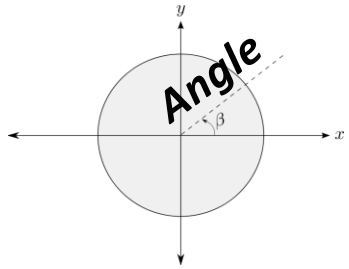
FORTH FOUNDATION FOR RESEARCH AND TECHNOLOGY - HELLAS
Université Paris Cité
institut universitaire de France
Université Mohammed V de Rabat

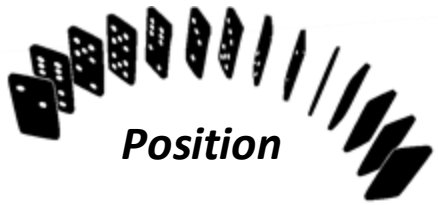
Data Series

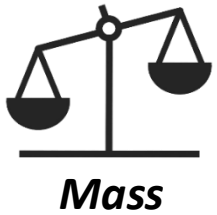
- A data series is a collection of points ordered over a dimension

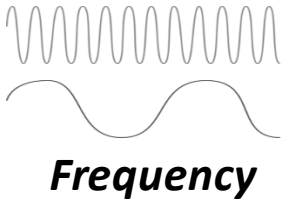










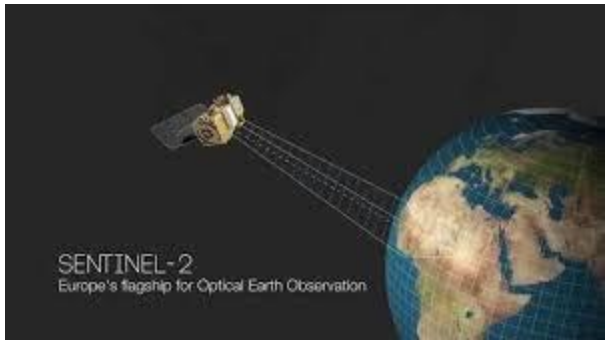


Data Series are Massive

≈ 500 ZB per year

≈ 130 TB

> 40 PB per day



> 5 TB per day

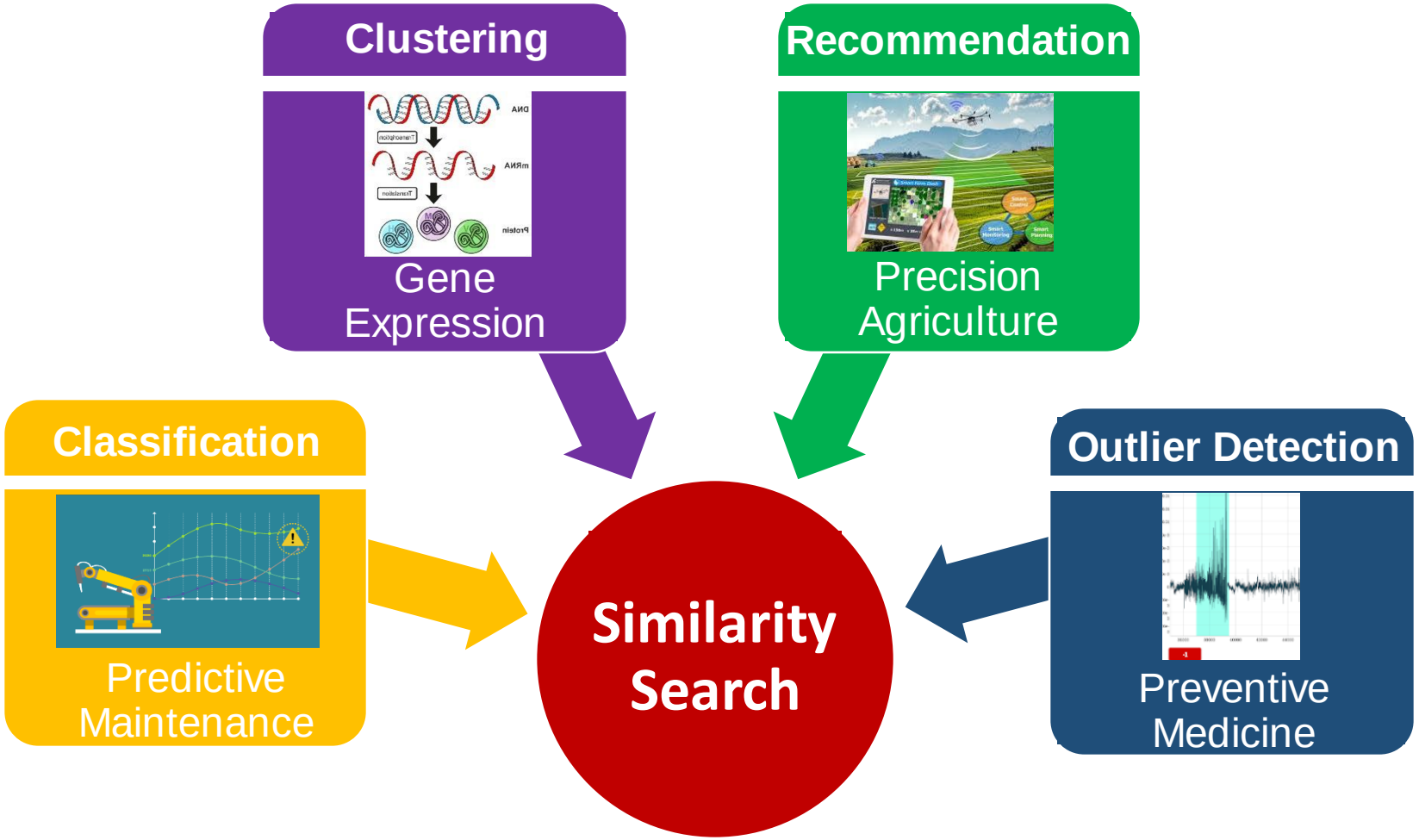
facebook

> 500 TB per day

1 PB = 1 thousand TB

1 ZB = 1 billion TB

Similarity search is a key bottleneck in analytics tasks



Similarity Search – Nearest Neighbor Problem

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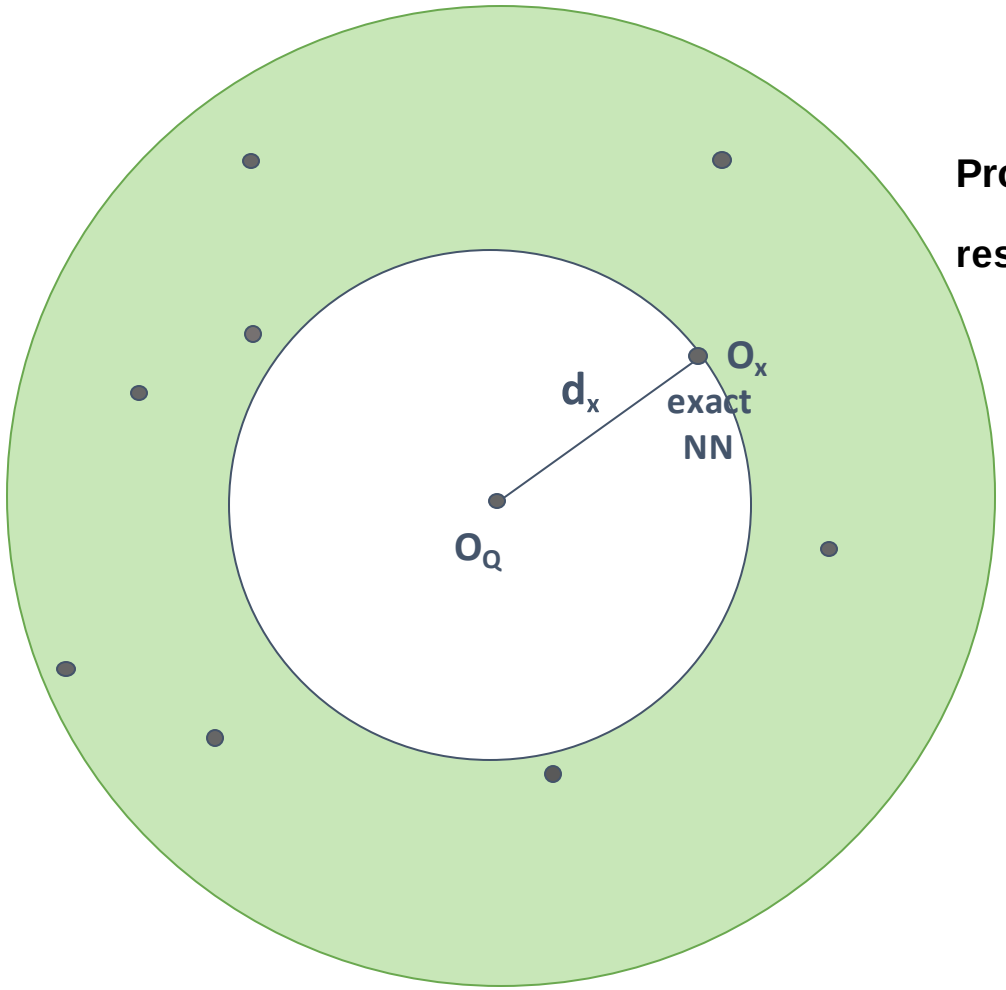

Prob($d_x = \min\{d_i\}$) = 1

result is exact NN

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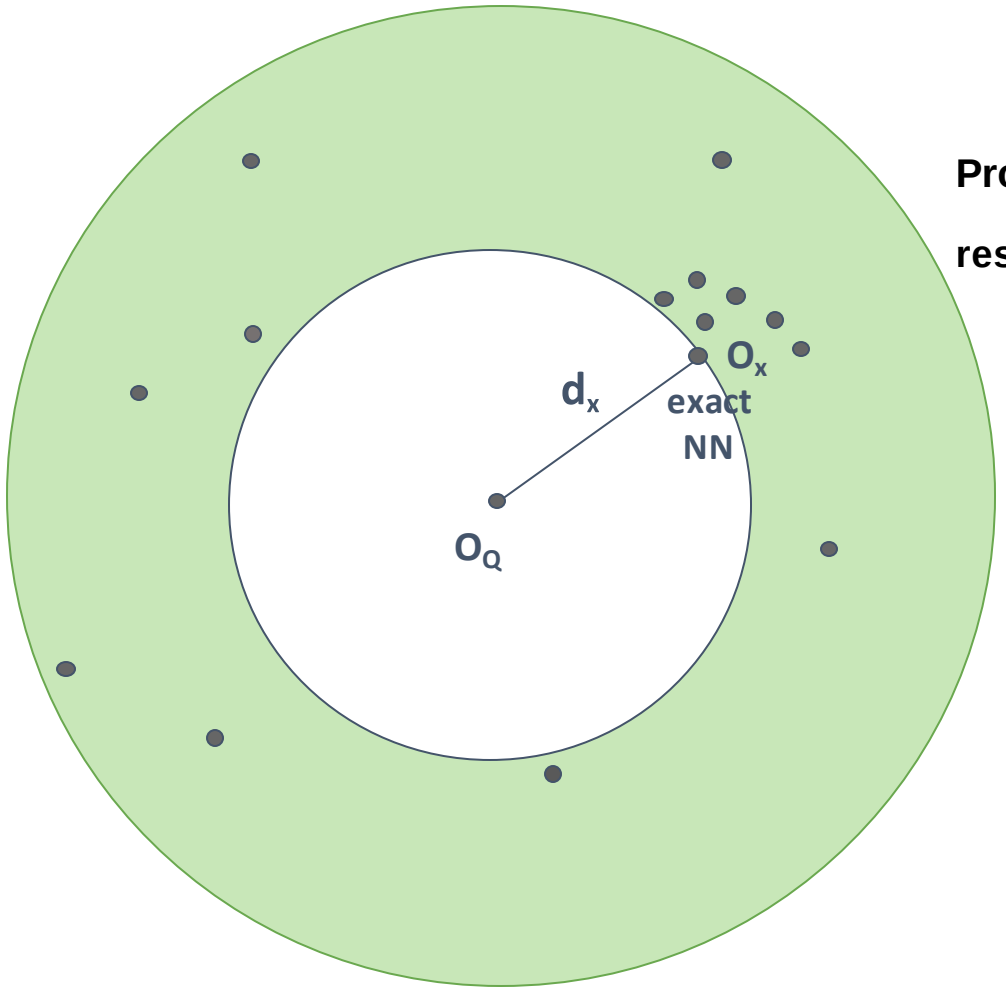

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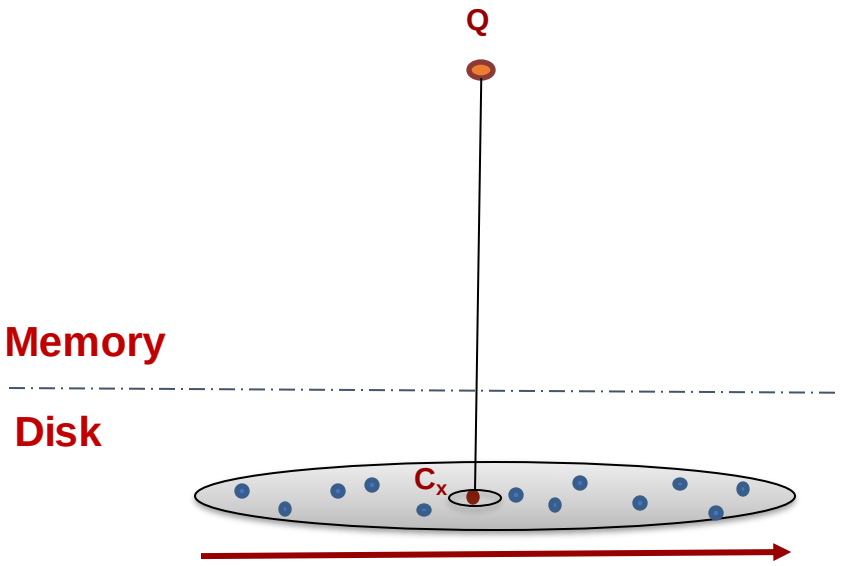
Exact Similarity Search Methods



Q is compared to each raw candidate in the dataset before returning the answer $\mathbf{C_x}$

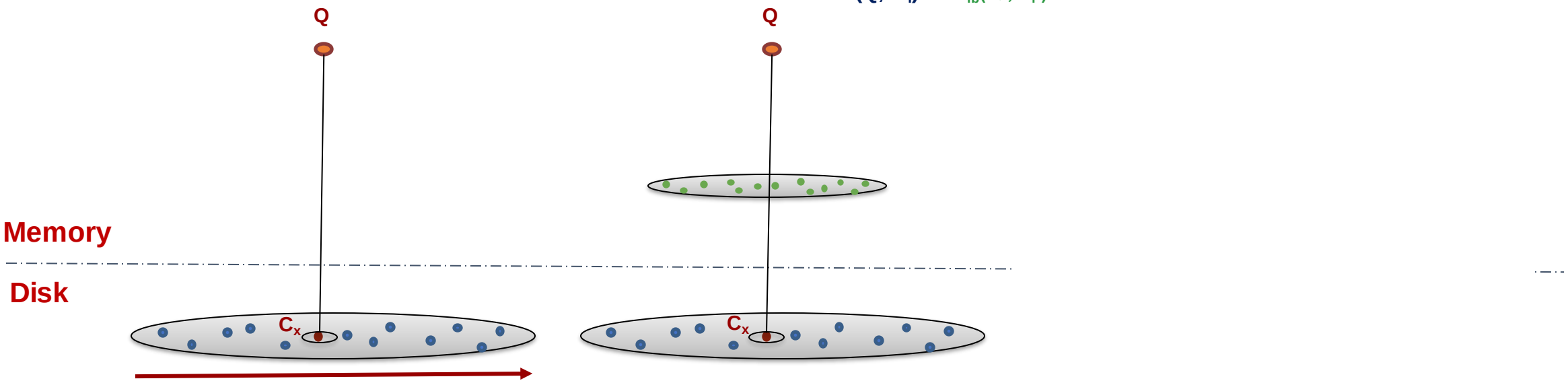
(a) Serial scan

Answering a similarity search query using different access paths

Exact Similarity Search Methods

lower-bounding (lb) property:

$d(Q, C_i) >= d_{lb}(Q', C_i')$



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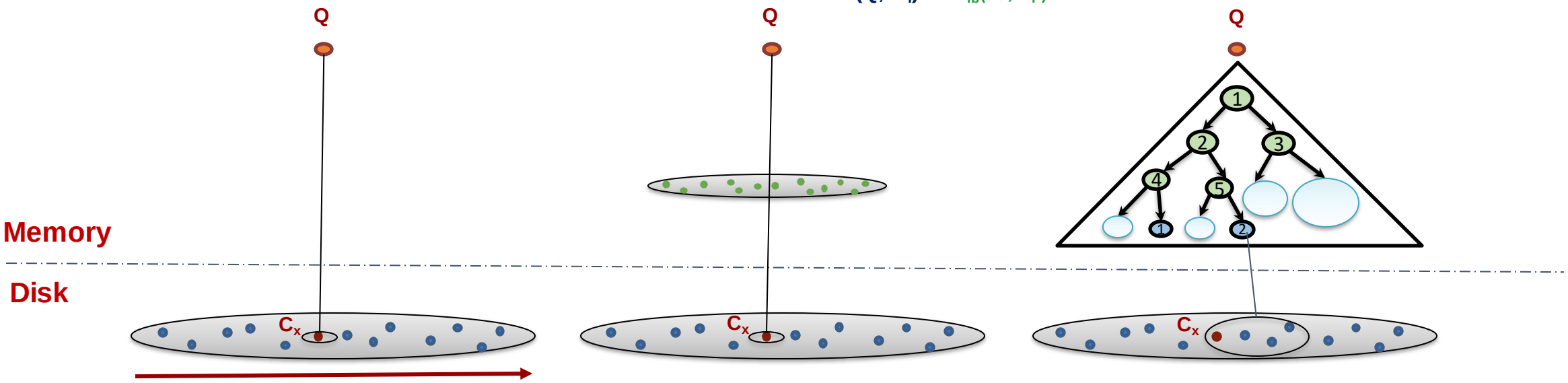

Q is compared to each raw candidate in the dataset before returning the answer C_x

Q is compared to a raw candidate only if its summary cannot be pruned

Q is compared to a raw candidate only if the subtree to which it belongs cannot be pruned

(a) Serial scan

(b) Skip-sequential scan

(c) Tree-based index

Answering a similarity search query using different access paths

Data Series Summarizations

V = [-1.5, -0.5, 0.5, 1.5, 2.5, 1.5, 2, 2.6]

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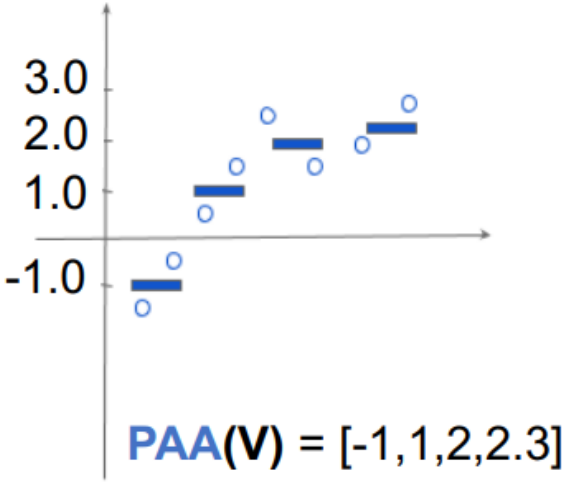

(a) PAA

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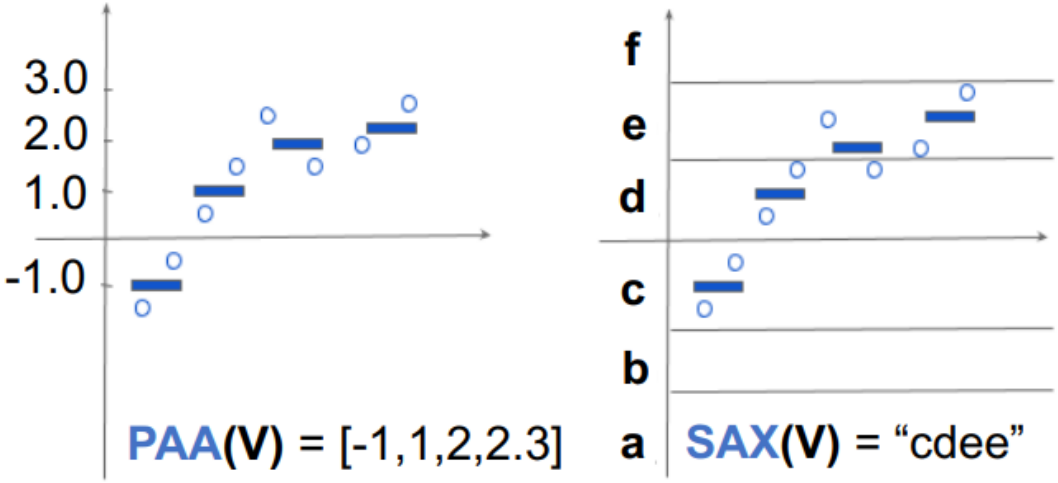

(a) PAA **(b) iSAX**

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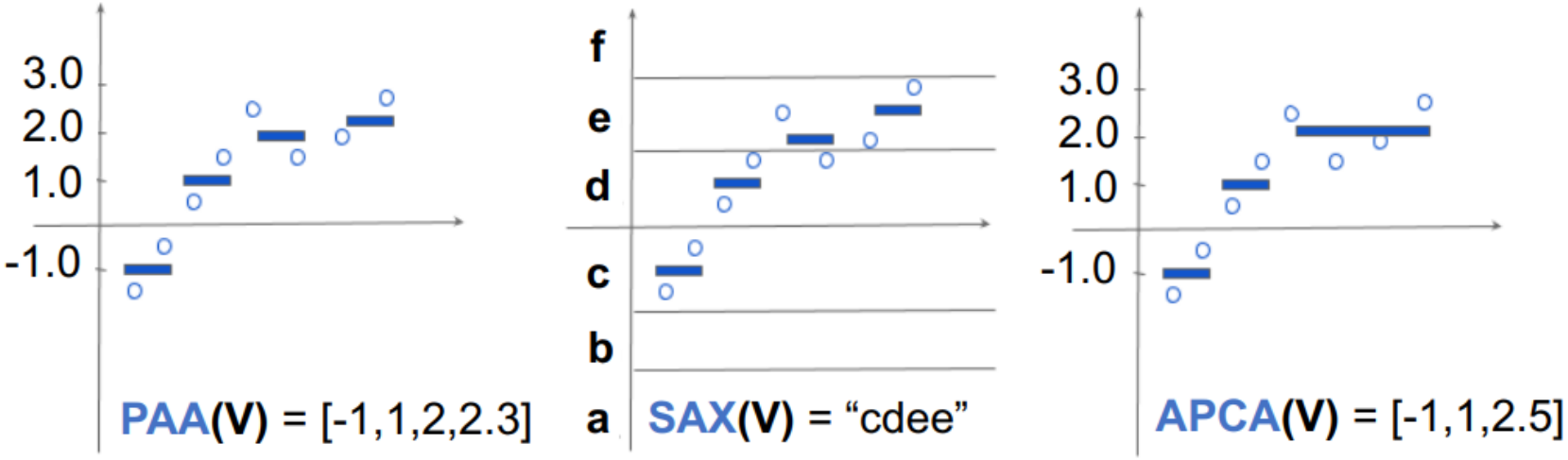

(a) PAA (b) iSAX (c) APCA

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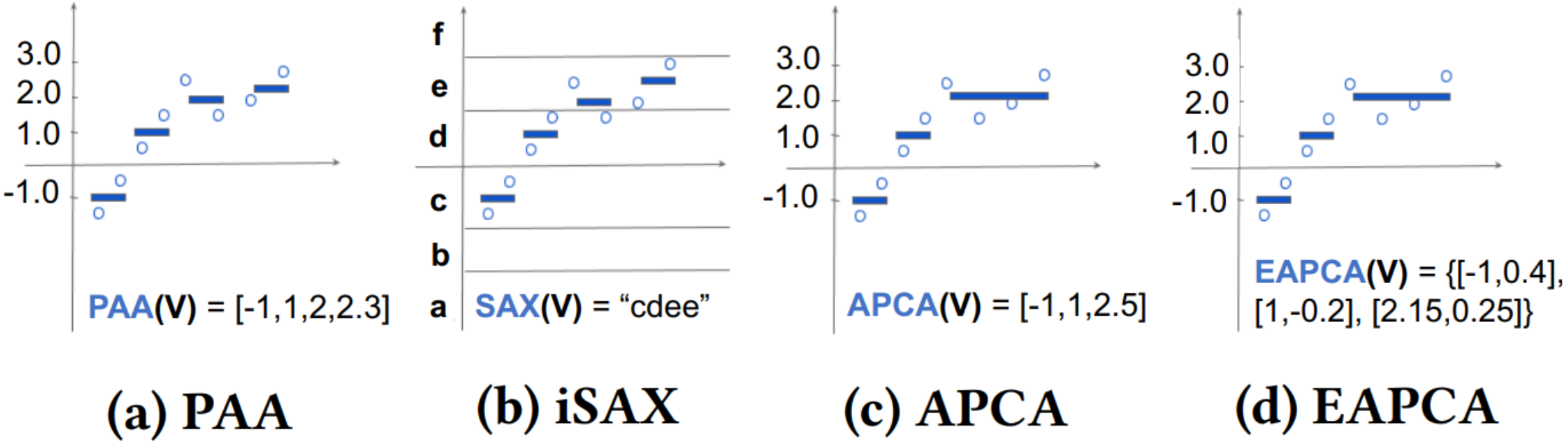

(a) PAA (b) iSAX (c) APCA (d) EAPCA

Key Challenges

- No existing approach wins across the board [1]:

[1] Echihabi et al. The Lernaean Hydra of Data Series Similarity Search: An Experimental Evaluation of the State of the Art. PVLDB 12(2): 112-127 (2018)

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- Using the same parallelization strategy for all query workloads can lead to performance degradation

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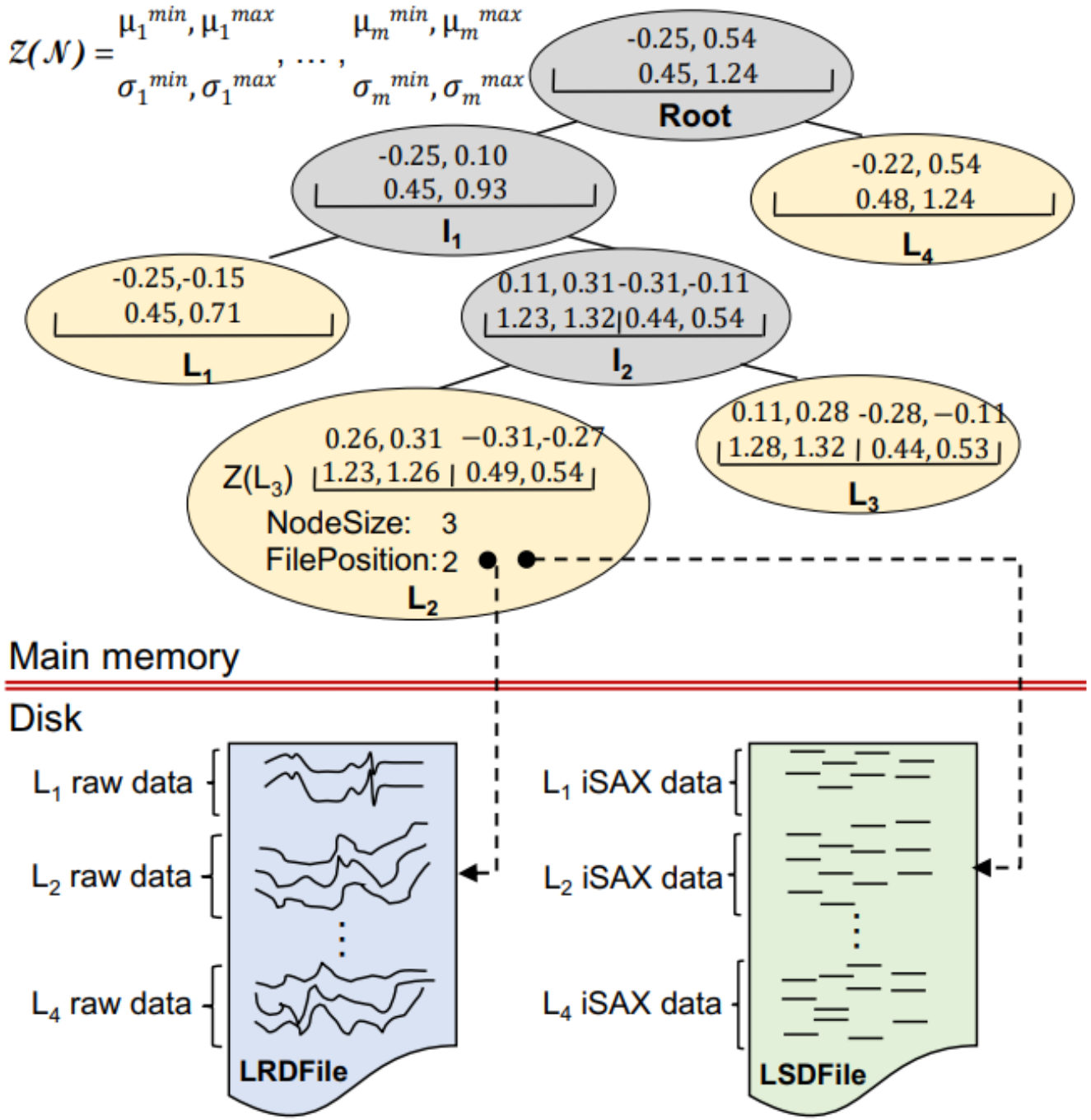
  - Performs up to one order of magnitude faster than the best competitor (which is not always the same)

The Hercules Tree



Hercules's Index Building

(i) Read Phase: coordinator reads data

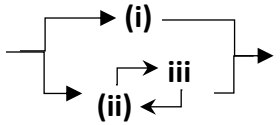
(ii) Insert Phase: insertWorkers insert series in index tree

(iii) Flush Phase: If memory full, insertions halt until flushCoordinator flushes HBuffer

Main memory

Disk

phase (i) executes in parallel with phases (ii) and (iii)



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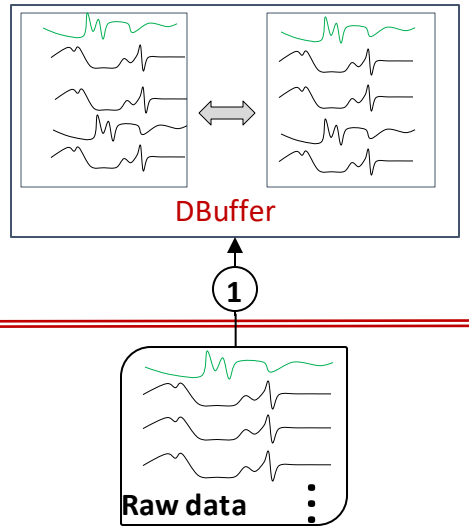

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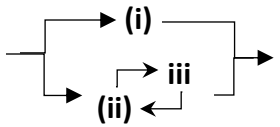

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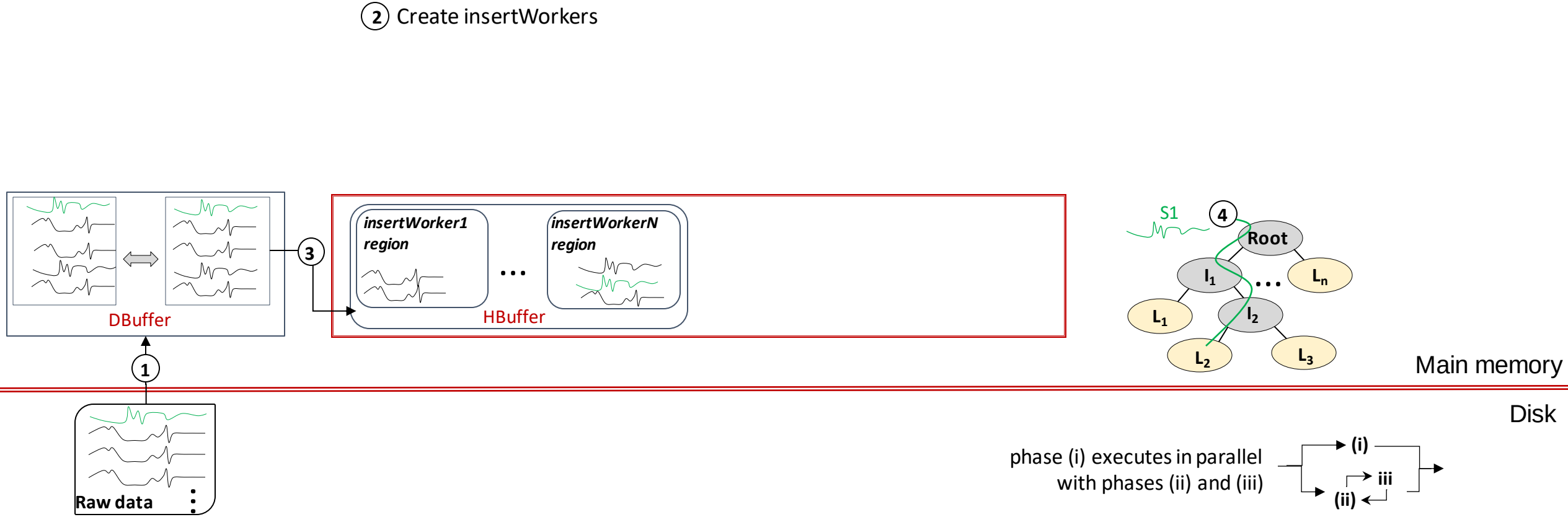

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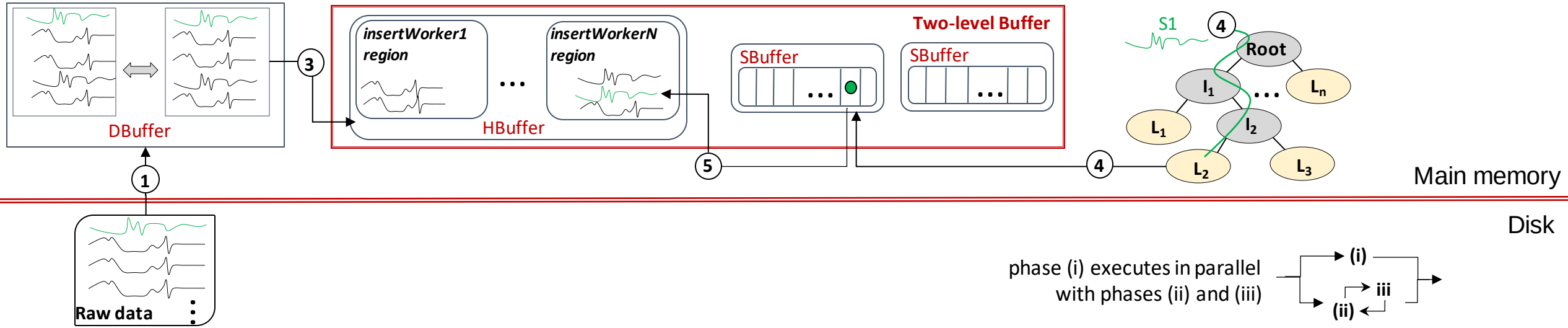

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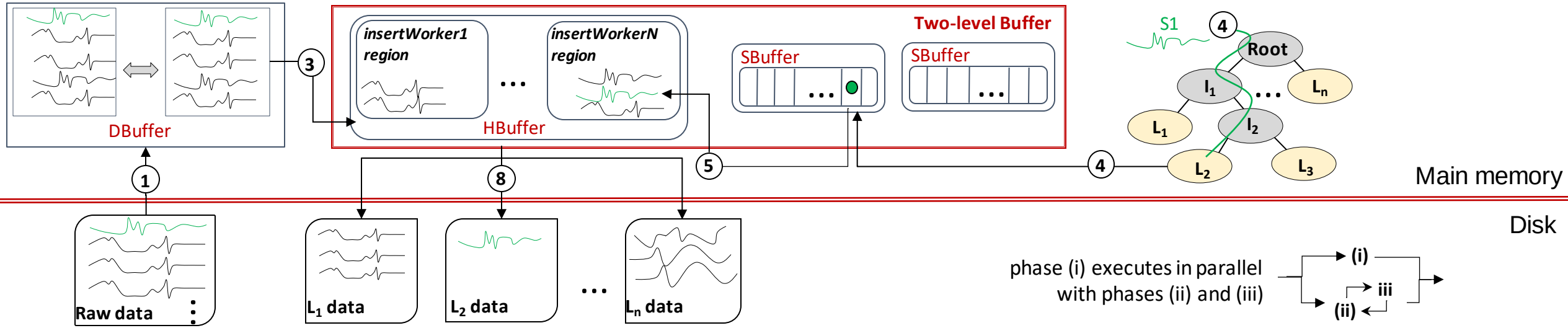
(2) Create insertWorkers

(6) Issue flushOrder if Ψ regions of HBuffer are full

(7) If flushOrder, halt insertions until flushCoordinator is done flushing



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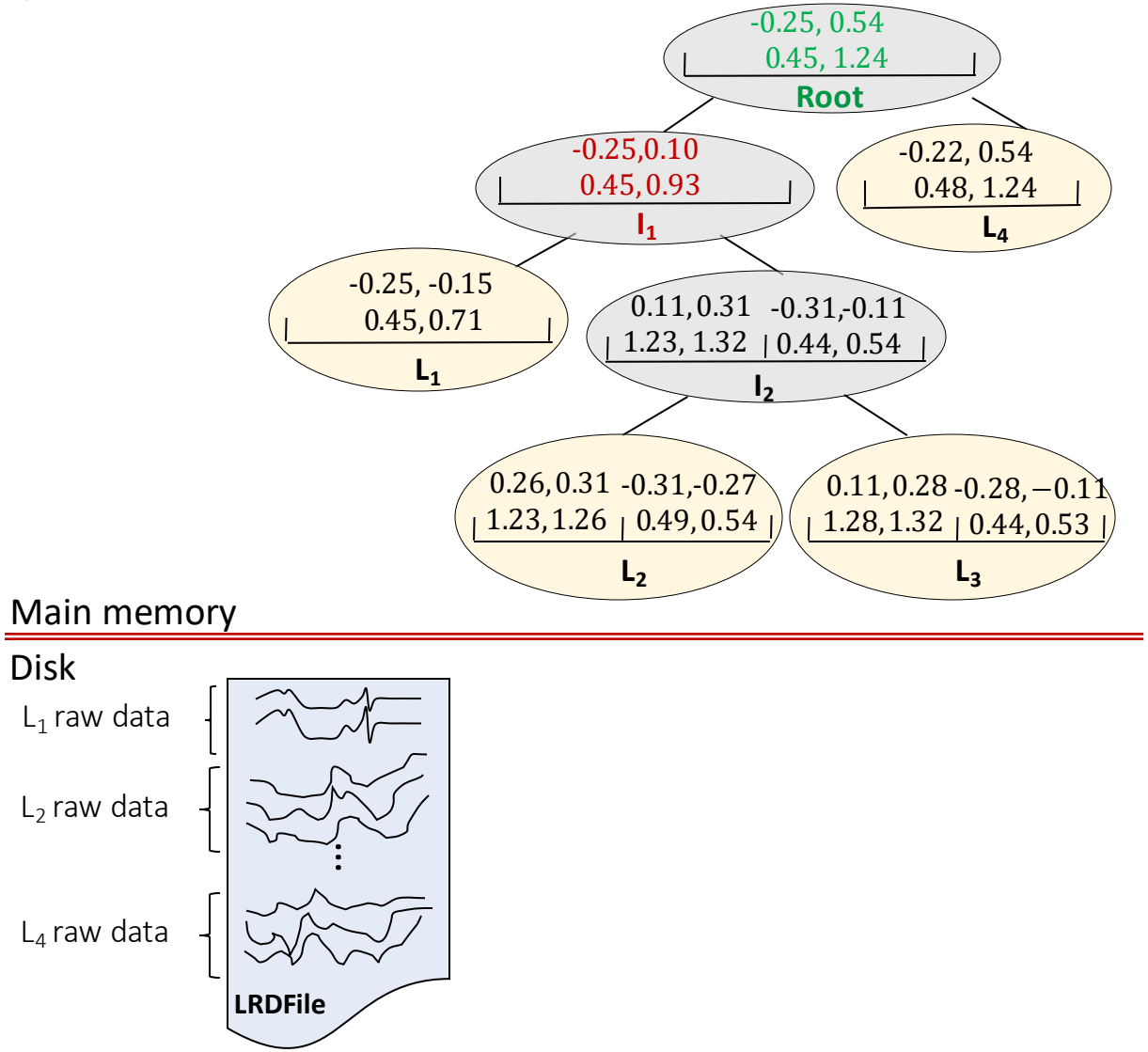

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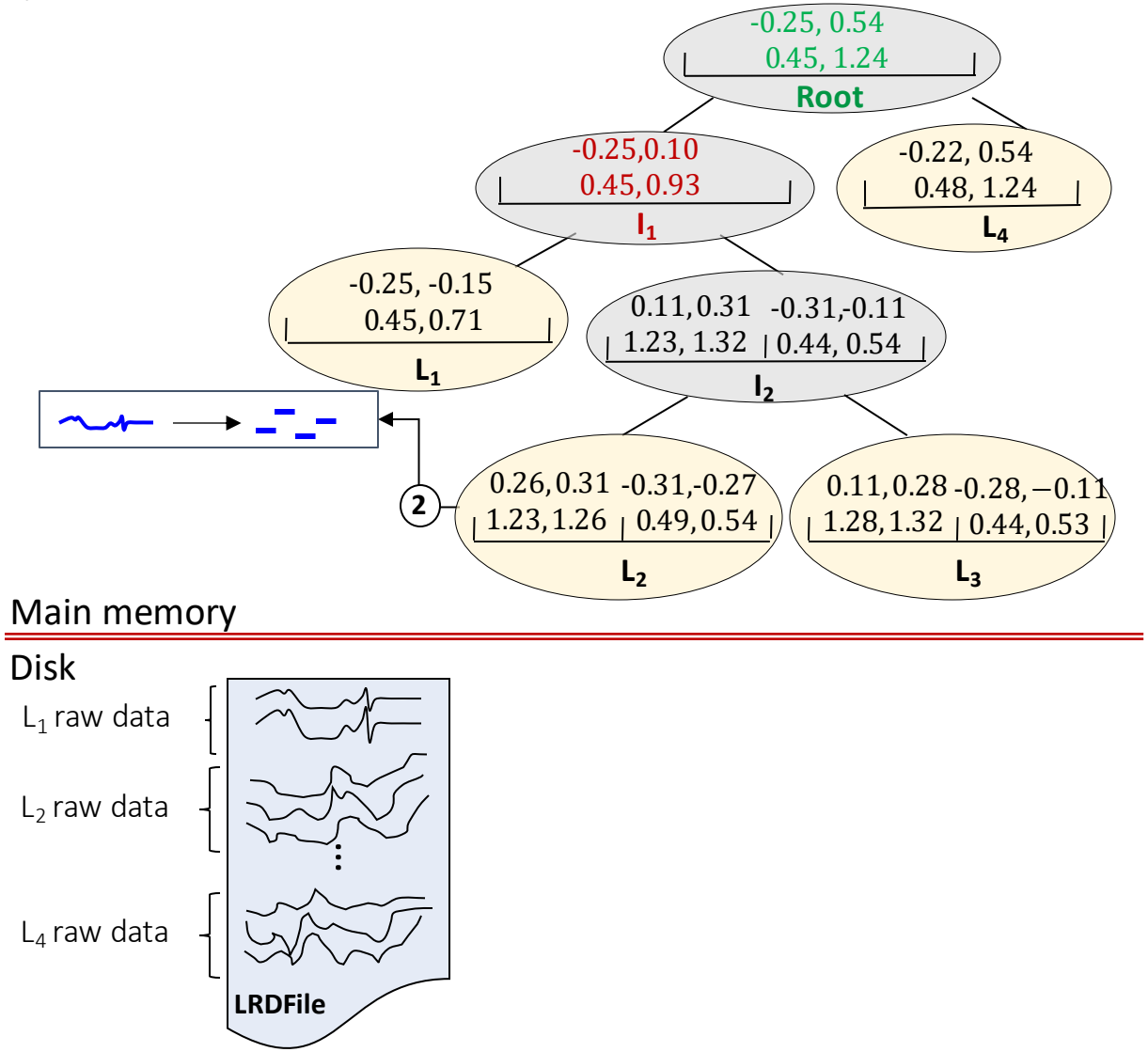

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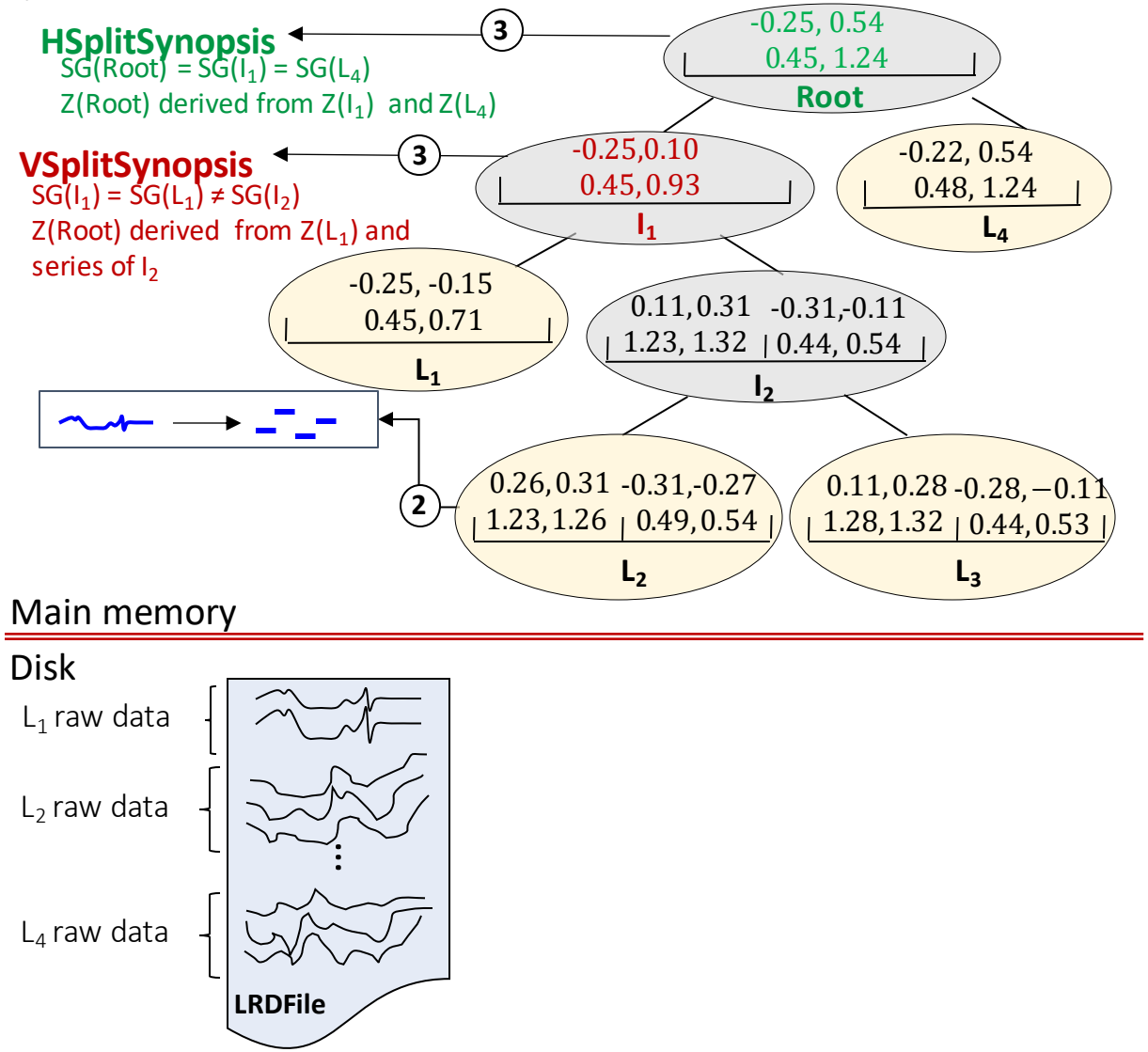

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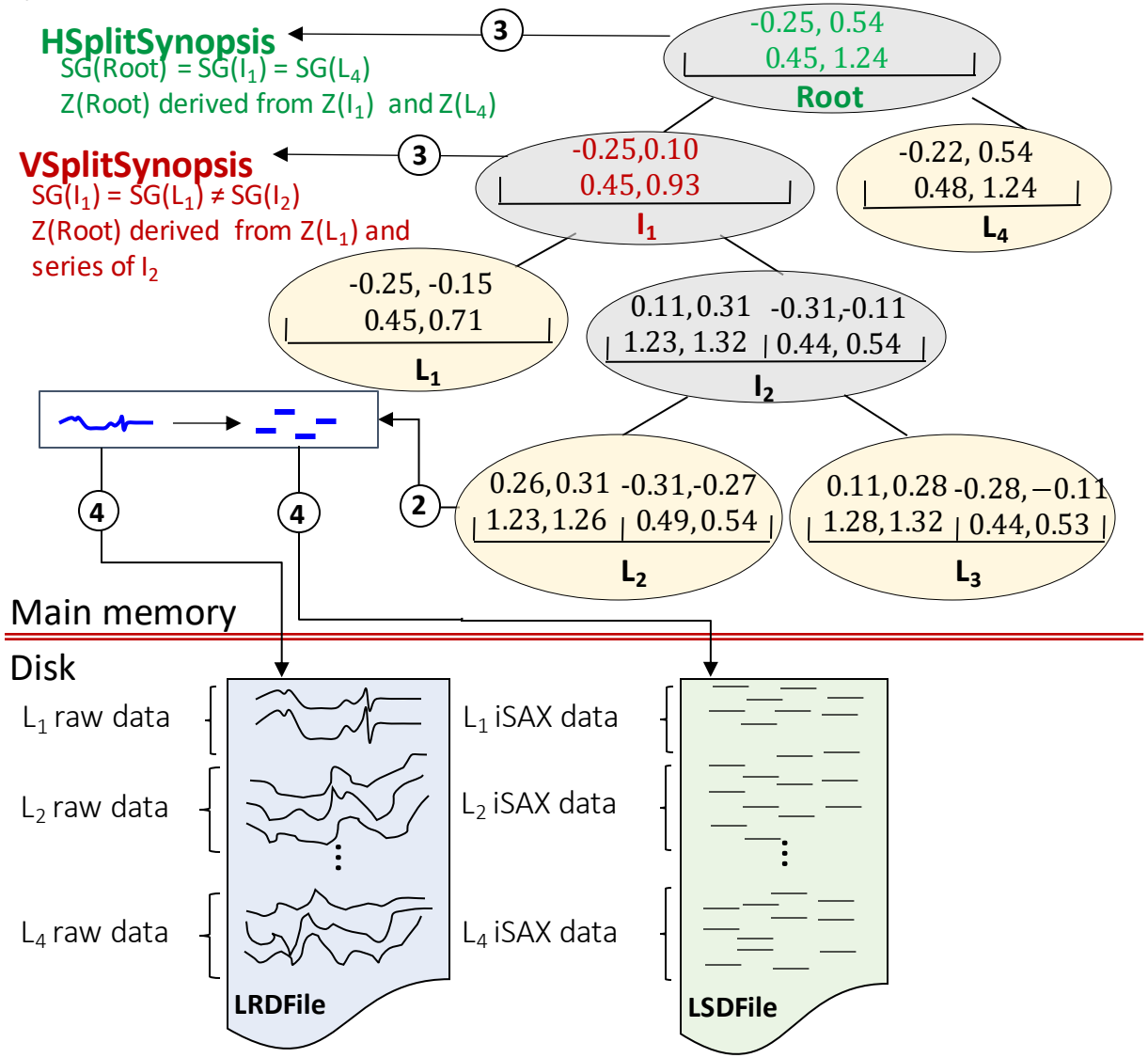

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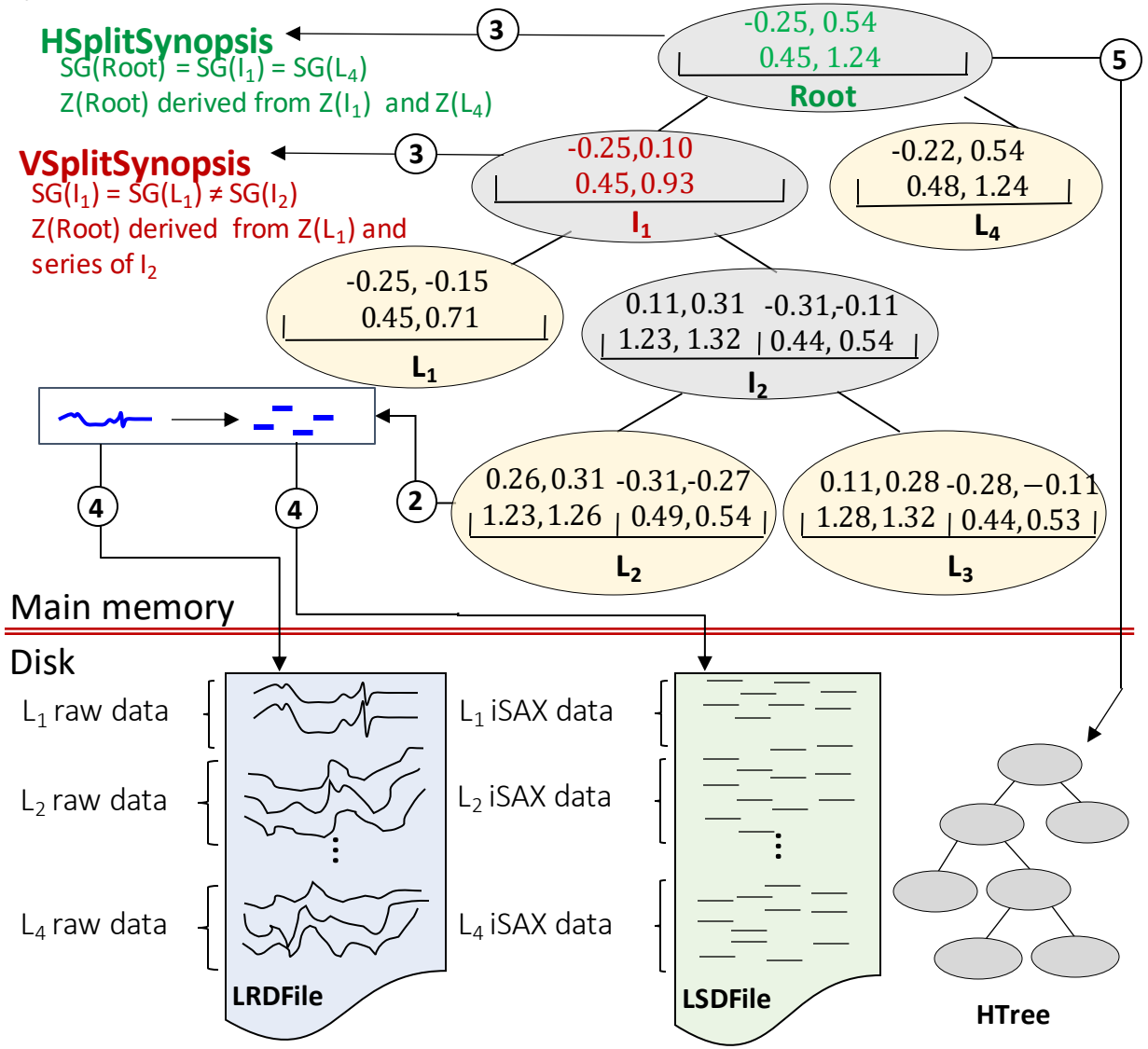

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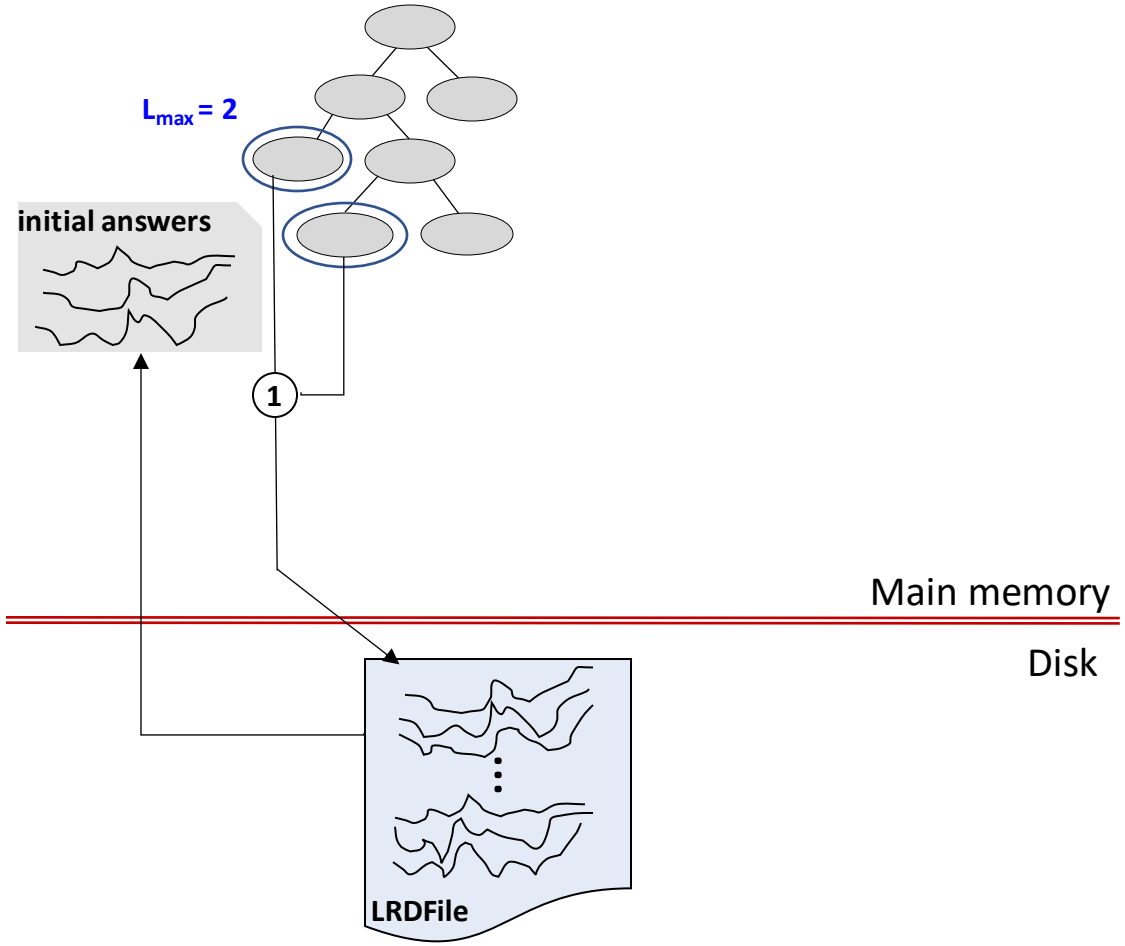

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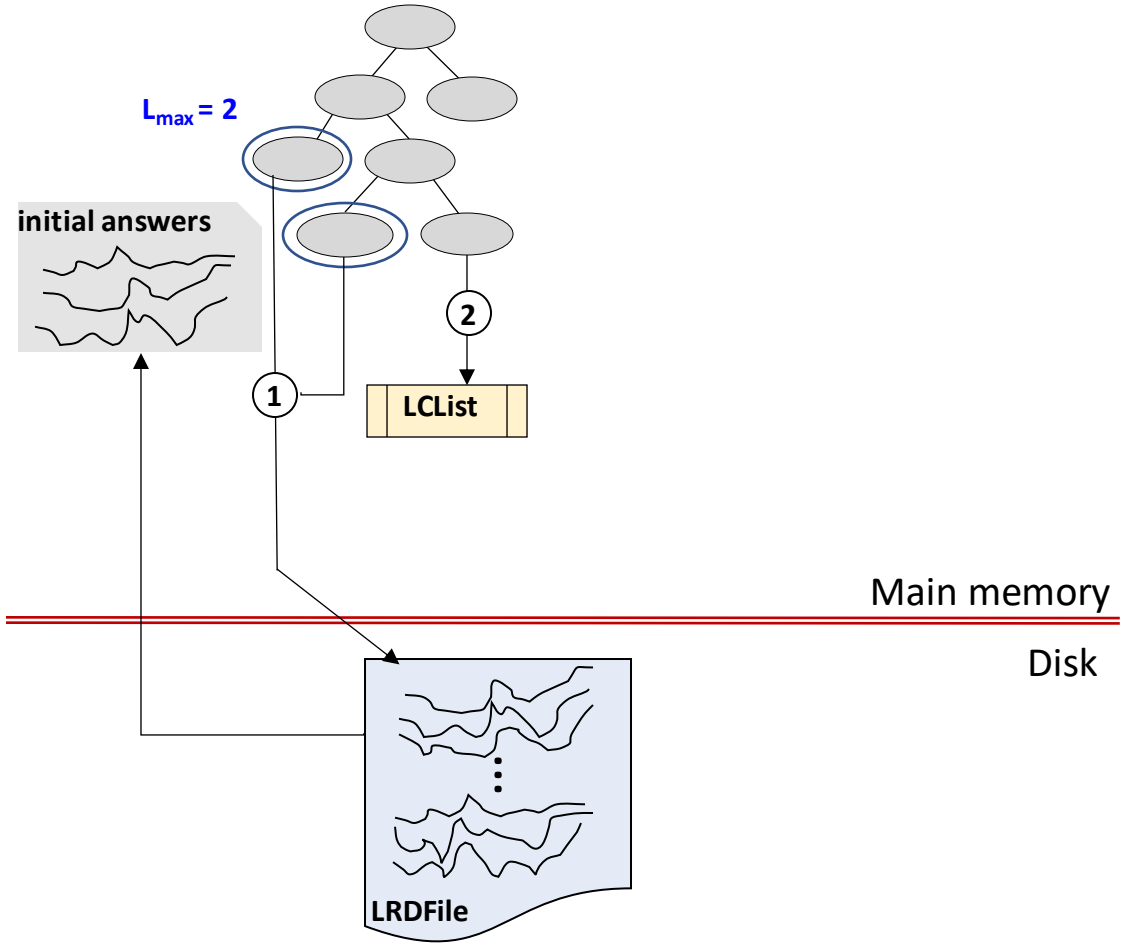

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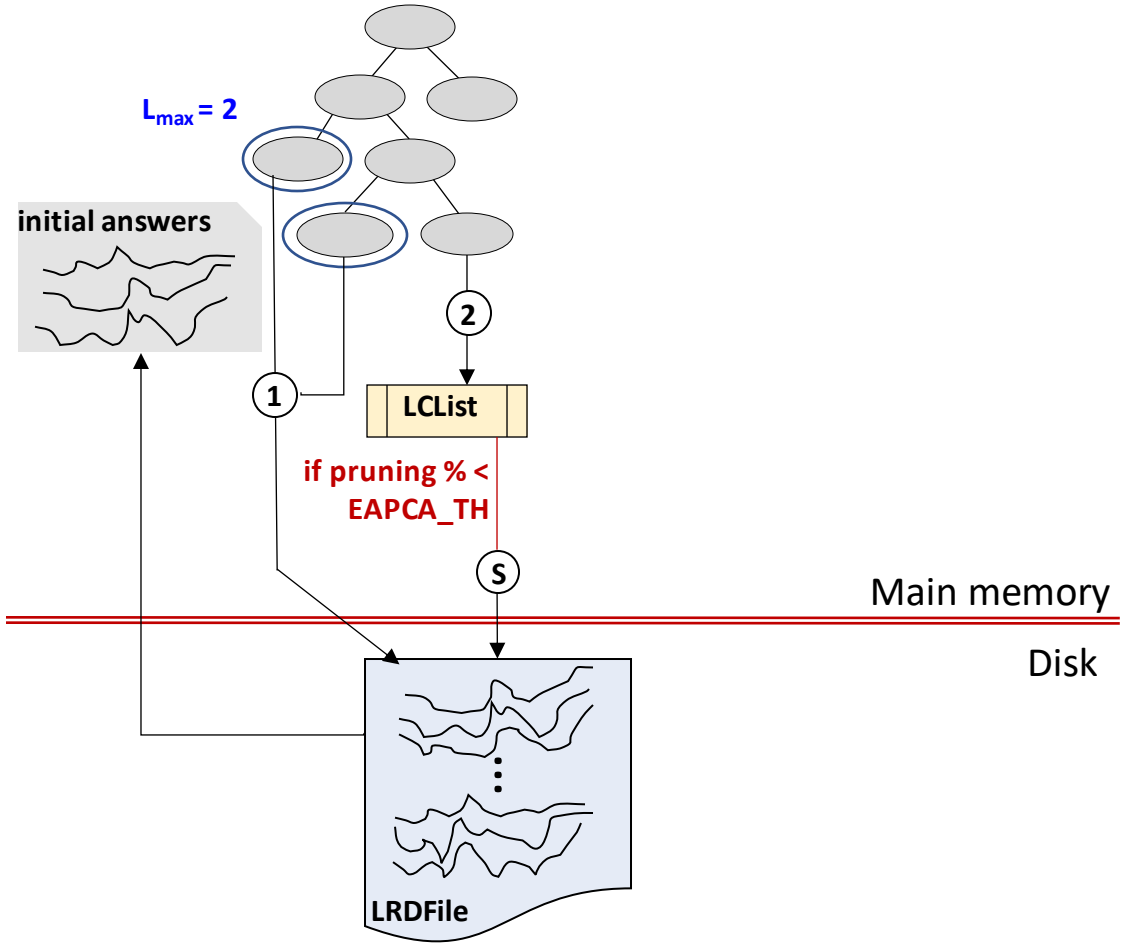

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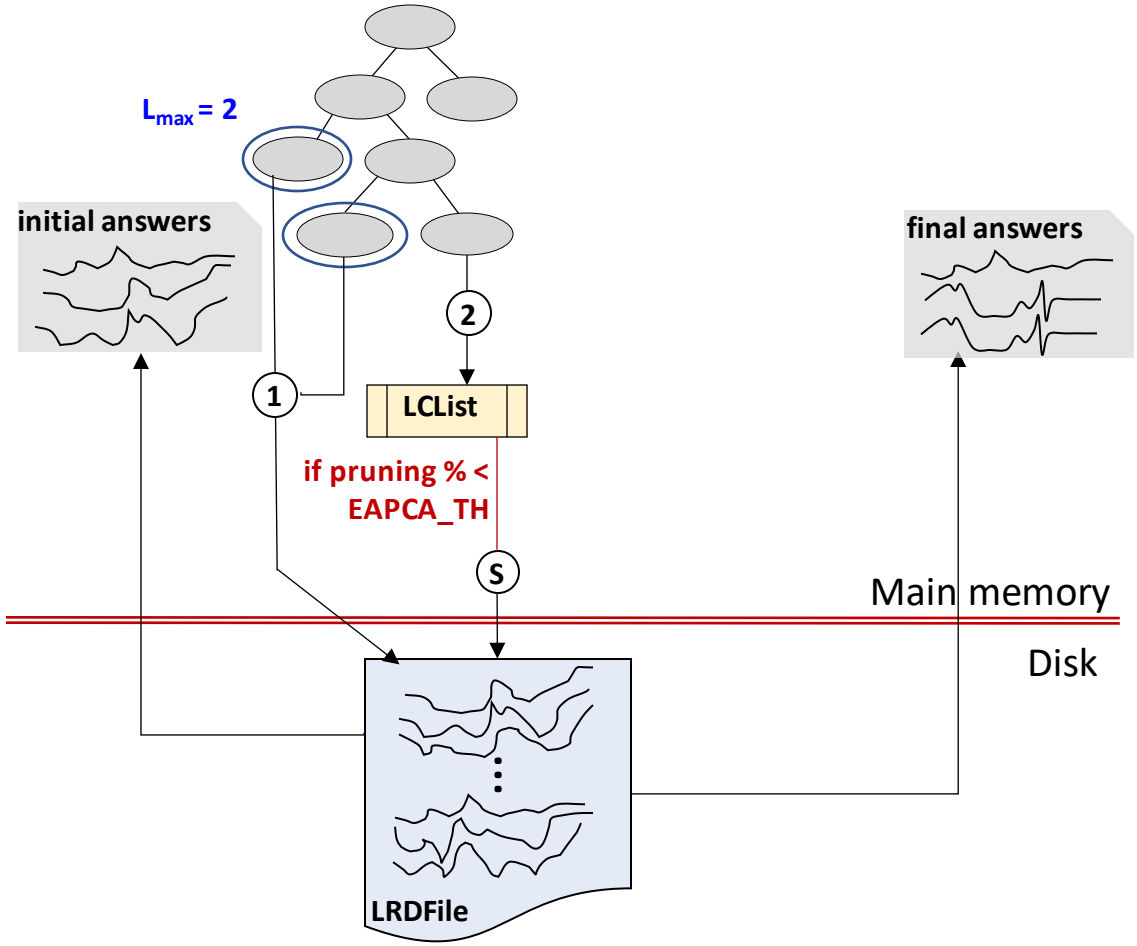

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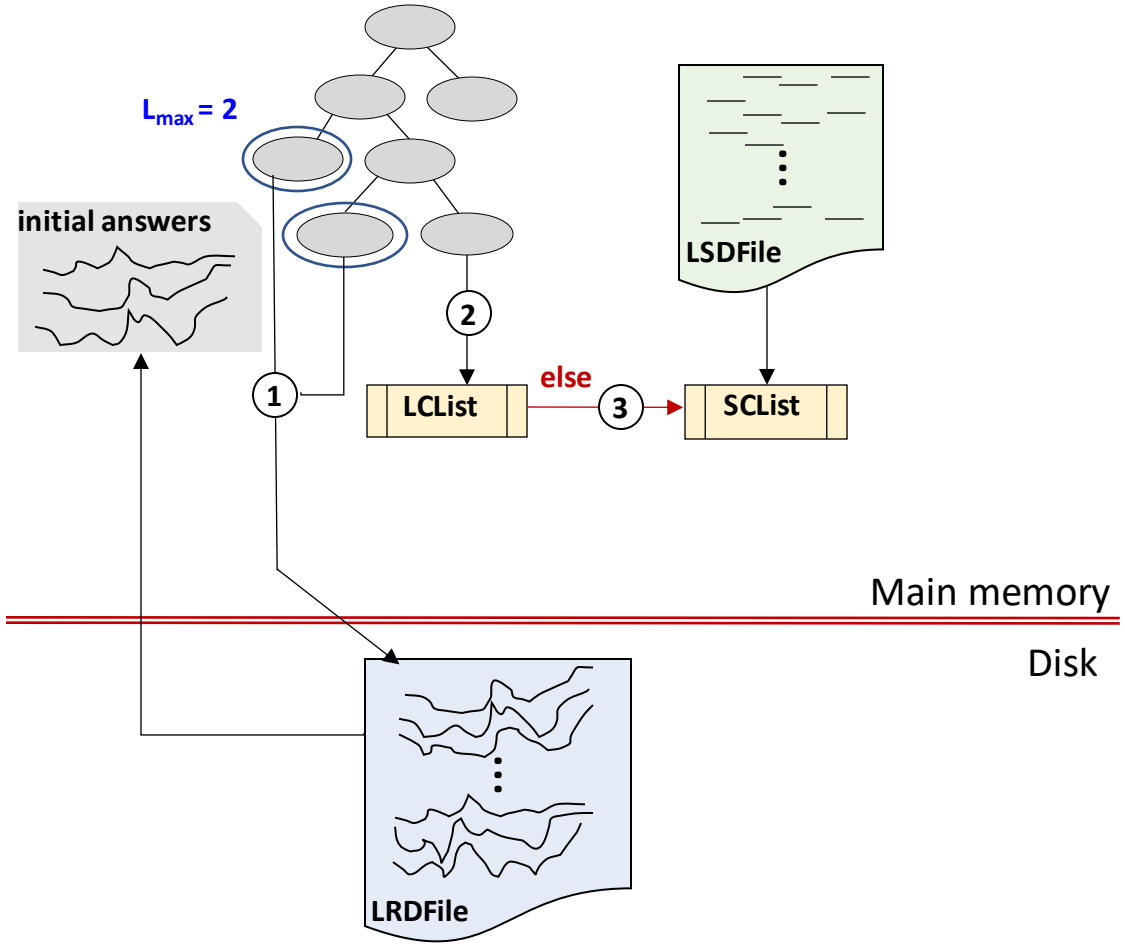

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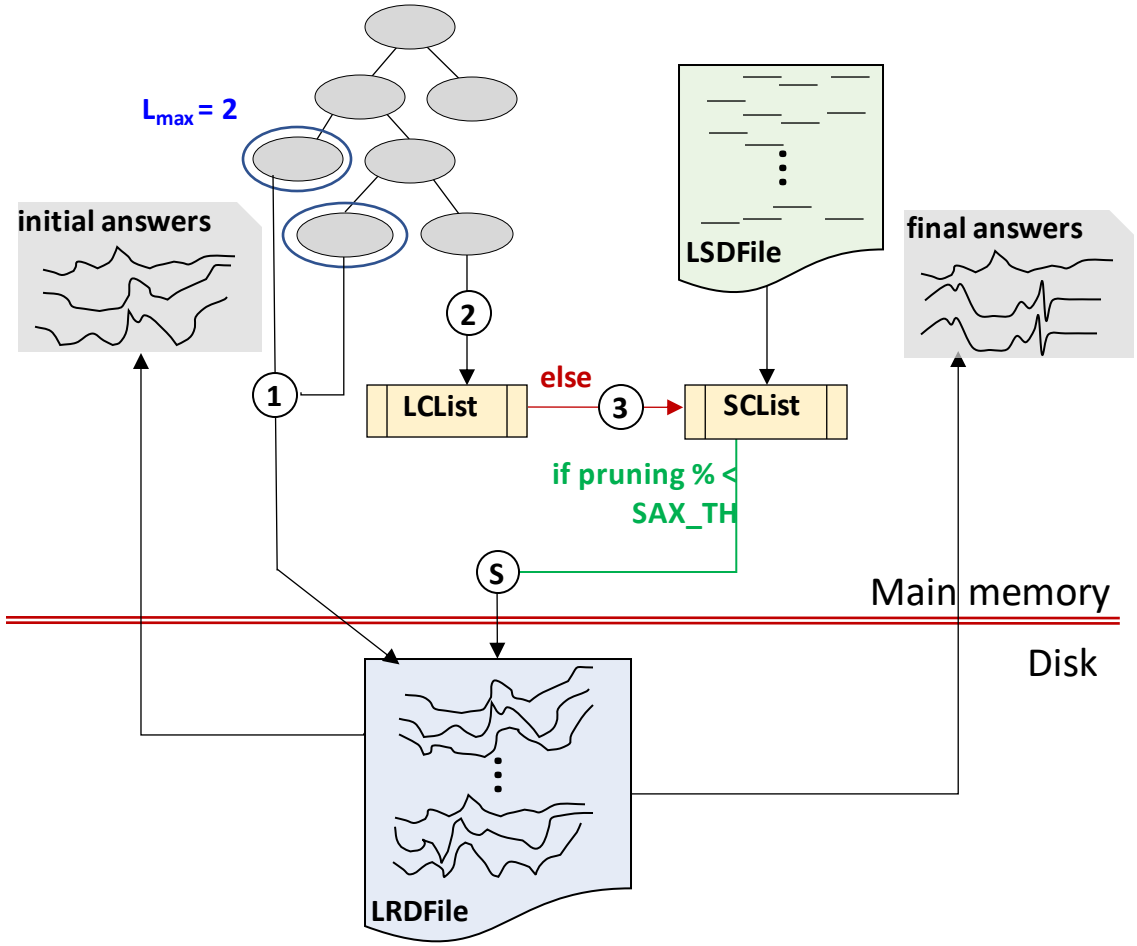

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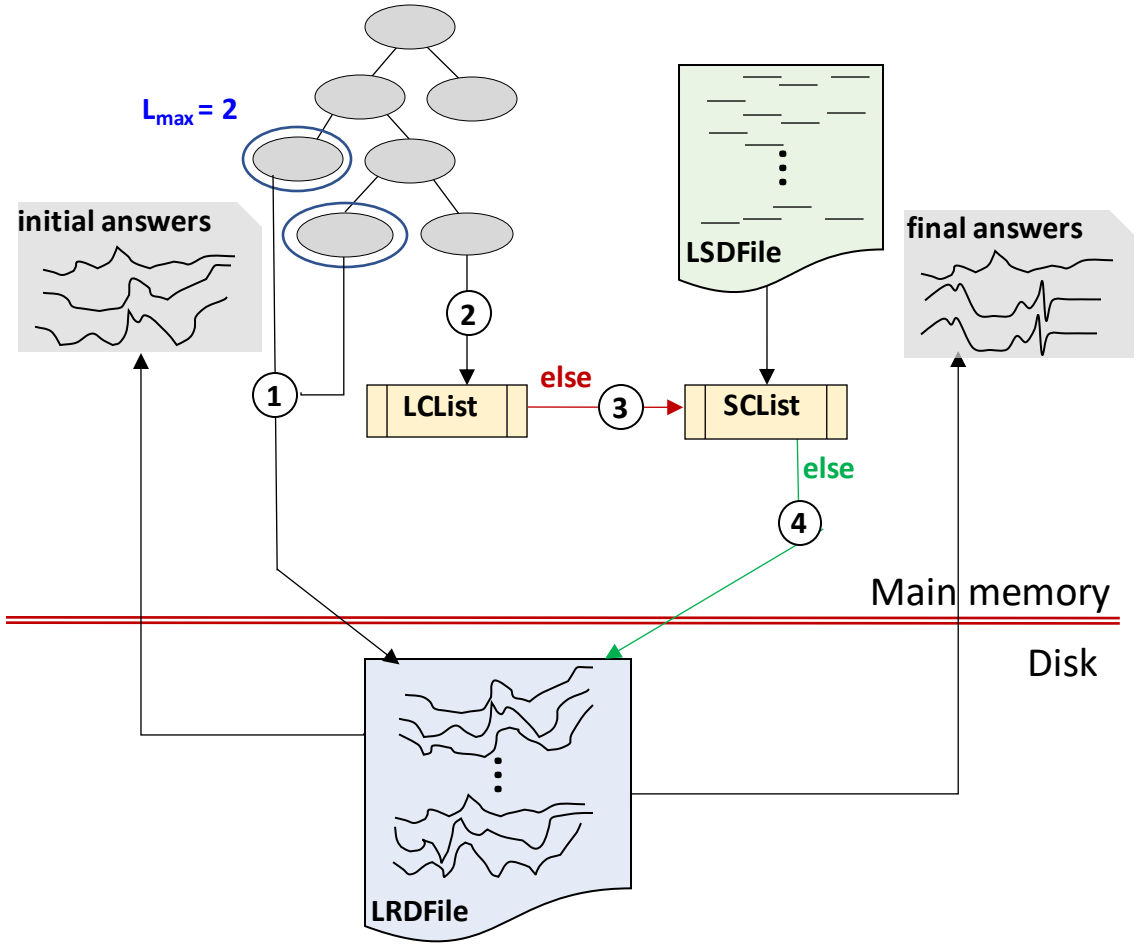
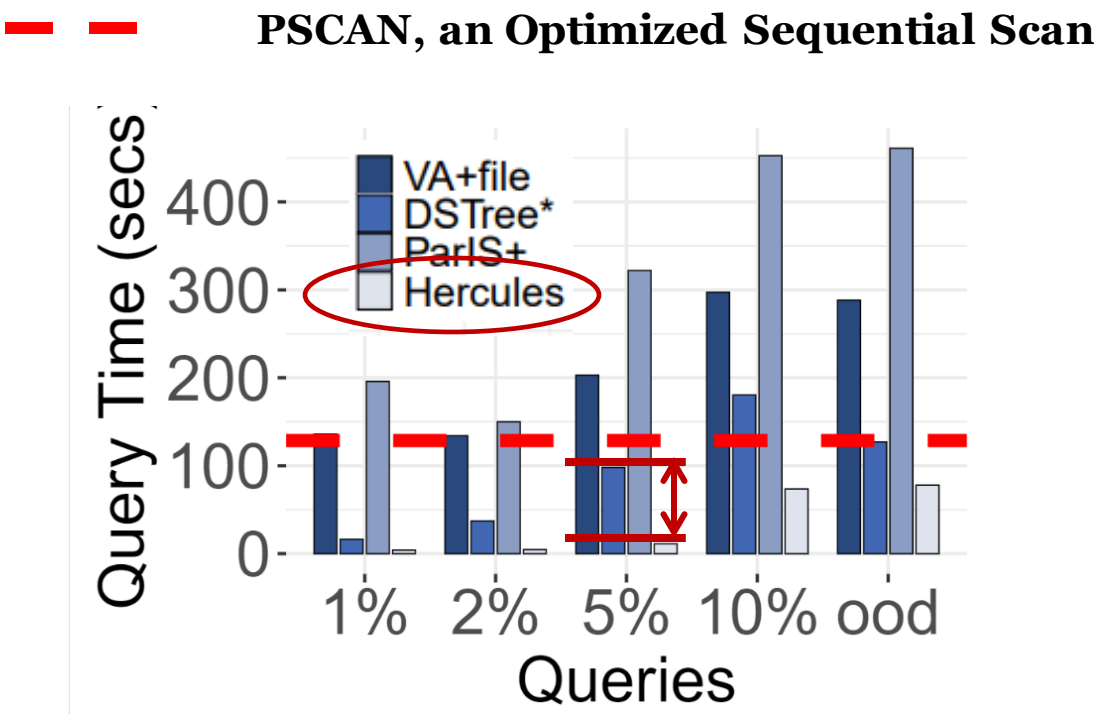


Experimental Setup

- Baselines:
  - VA+file
  - DSTree
  - ParIS+
  - PSCAN: optimized parallel sequential scan
- Datasets
  - Synthetic (25GB to 1.5TB and 128-16384 dimensions) and 3 real (100 GB) from different domains
- Exact Query Workloads
  - 100 – 10,000 kNN queries (k in [1,100])
  - Queries of increasing difficulty
- Performance measures
  - Time, pruning
- Procedure:
  - Parametrization
  - Scalability Experiments
  - Ablation Study

Main Results

- Scalability with query difficulty (Deep100GB)



10x faster than best competitor

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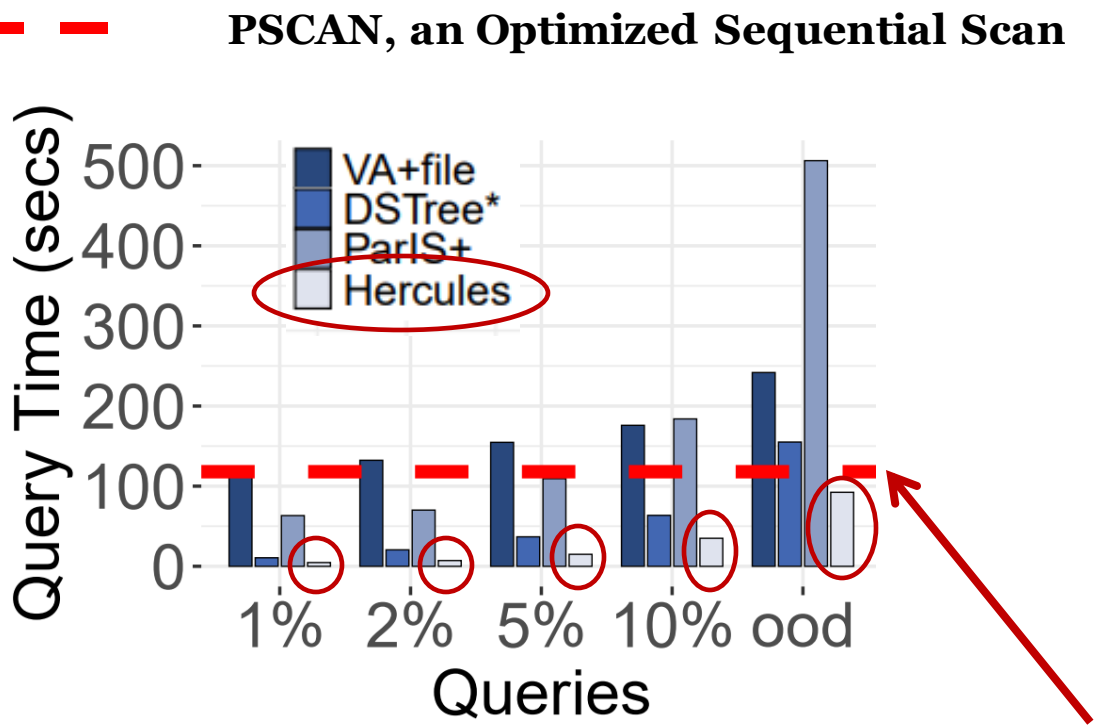
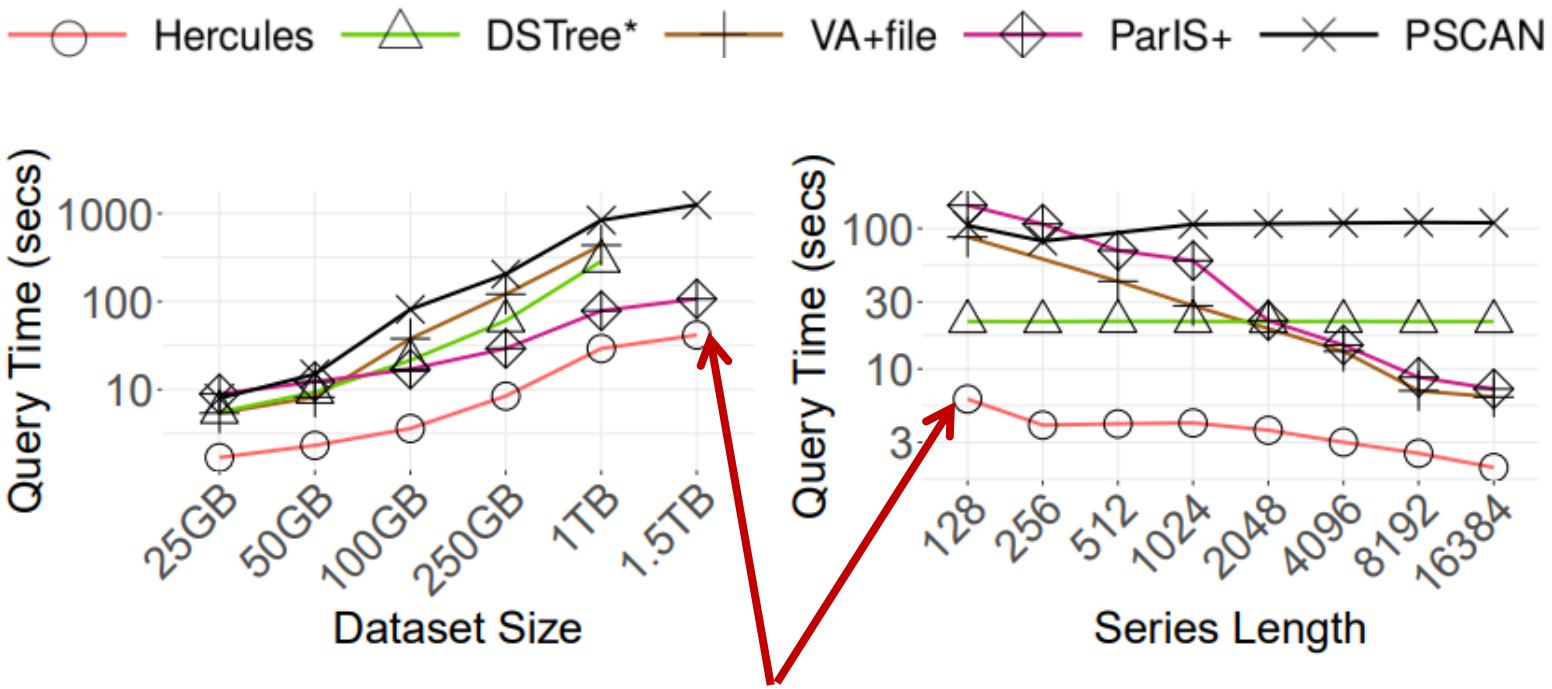


Only index always better than optimized sequential scan

Main Results

- Scalability with dataset size and series length (Synthetic)



Hercules best overall

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